

- 6** A car with a power output of 90 kW can travel at a maximum speed of 55 m s^{-1} along a straight horizontal road. The drag force on the car is directly proportional to the square of its speed.

What is the maximum speed of the car, if the power output of the car is reduced to 50 kW while travelling on the same road?

A 31 m s^{-1}

B 41 m s^{-1}

C 45 m s^{-1}

D 67 m s^{-1}

- 7** An object of mass 1.5 kg slides down a rough plane inclined at 10° above the horizontal